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AI-Enabled Unconfined Compressive Strength (UCS) Real-Time Prediction Utilizing Logging-While-Drilling Measurements

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The quality and applicability of the training data used to create predictive models have a major impact on how well data-driven drilling decisions work. The oil and gas sector has been using machine learning (ML) techniques more and more in recent years to glean actionable insights from massive databases of historical and real-time drilling data. Drilling operations planning and execution are being redefined by the capacity to train intelligent systems that can recognize patterns and predict subsurface behavior. The use of logging-while-drilling (LWD) data streams is essential to this evolution because they offer high-frequency information that, when properly utilized, can be crucial in lowering non-productive time (NPT), improving drilling efficiency, and boosting operational safety.

The Unconfined Compressive Strength (UCS), which measures a formation's resistance to mechanical failure, is one of the most important rock characteristics affecting drilling performance. Important drilling choices like the ideal Weight on Bit (WOB), rotary speed, and bit type are all directly influenced by UCS. Engineers can react dynamically to changes in the formation by adjusting parameters to either drill more aggressively through soft zones or reduce mechanical stress in harder, abrasive intervals when they have a solid understanding of UCS in real time. Continuous UCS insight becomes not only valuable but crucial in abrasive, heterogeneous sandstone formations, where sudden variations in rock strength can result in excessive bit wear or ineffective drilling.

However, real-time access to UCS data has historically been limited. Conventional methods rely on core samples or post-drilling analysis of wireline logs, both of which suffer from delayed availability and limited spatial resolution. These gaps prevent timely optimization and expose operations to avoidable risks. By contrast, a system capable of continuously estimating UCS from surface-acquired or downhole LWD data represents a step-change in drilling intelligence.

Introduction

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This study presents a machine learning-based workflow that bridges this gap by predicting sonic slowness (DTC) in real time from standard LWD measurements and subsequently estimating UCS using established empirical equations. The model is deployed on an autonomous petrophysical platform that delivers continuous UCS predictions to the drilling team during operations. This enables near-instantaneous adjustments to drilling parameters based on predicted rock strength, allowing for more precise control of the drilling process.

The implications of automating UCS prediction using machine learning are substantial. Real-time feedback empowers drilling engineers to proactively adjust drilling strategies, minimize impact on the drill bit at higher UCS rocks, extend bit life, and avoid unnecessary trips. From a broader operational standpoint, the approach contributes to derisking drilling operations by minimizing the time spent drilling, optimizing the number of drill bits used, and saving rig time. Financially, it aids in reducing rig time, lowering equipment costs, and improving the overall efficiency of drilling. As the industry continues to embrace digital transformation, intelligent, automated workflows like the one presented here are poised to become integral components of modern drilling operations.

Methods, Procedures, and Processes

The proposed workflow integrates machine learning, empirical geomechanics, and real-time operations monitoring to predict Unconfined Compressive Strength (UCS) and optimize drilling parameters dynamically. The methodology is structured into three interconnected phases: the Planning Phase, the Monitoring and Model Update Phase, and the Decision-Making and Optimization Phase.

Planning Phase: UCS Modeling and Trajectory Preparation

The workflow begins in the planning stage by developing a predictive model for UCS based on machine learning and empirical relationships. A Random Forest algorithm was trained to predict compressional slowness (DTC) using common Logging-While-Drilling (LWD) logs, specifically Gamma Ray (GR), Bulk Density (RHOB), and Thermal Neutron Porosity. These logs are routinely collected and readily available in most drilling operations, adding no additional cost or data acquisition burden.

The training dataset was compiled from several wells within the same geological formation of interest, all of which included measured DTC logs. The model was trained on a subset of these wells, while validation was performed on a holdout set to avoid overfitting. Model performance was evaluated using Mean Squared Error (MSE), and the best-performing model was selected from an ensemble based on its predictive accuracy.

on the validation wells. For generalization assessment, the selected model was then deployed and tested on a separate group of wells not used during the training or validation stages.

Once DTC predictions are available, UCS is computed using McNally's (1987)¹ empirical relationship:

$$UCS = 1,430,000 \times e^{-0.035t}$$

Where UCS is in psi and t is the DTC (compressional travel time) in $\mu\text{s}/\text{ft}$.

The resulting UCS profile is then utilized during well trajectory planning. Zones of high and low UCS are analyzed to optimize the wellbore path and identify intervals of potential drilling risk. A drilling plan is prepared based on expected UCS variation, with recommended parameters including Weight on Bit (WOB), Rotary Speed, and Mud Flow Rate. An initial ROP model is also developed, incorporating UCS forecasts to set expectations for penetration rates and bit wear performance.

Monitoring and Model Update Phase: Real-Time Execution

During drilling, surface and downhole data are continuously collected. Surface data includes real-time measurements of ROP, WOB, torque, and mud flow rate, while LWD data streams provide updated GR, RHOB, and neutron porosity logs. These inputs are fed into the trained Random Forest model deployed on an autonomous petrophysics platform.

As drilling progresses, the platform continuously predicts DTC using real-time LWD logs. These DTC values are immediately converted to UCS estimates via the McNally equation. This stream of UCS predictions forms the basis for real-time rock strength profiling and performance evaluation.

By comparing actual ROP with expected ROP (based on UCS and drilling parameters), engineers can determine whether the drilling system is performing within optimal boundaries. This real-time loop allows for dynamic adaptation as formation properties change or if performance indicators (such as bit response or rate of penetration) diverge from expected trends.

Decision-Making and Optimization Phase: Performance Feedback and Adjustment

With a real-time UCS profile and continuously updated performance metrics, the drilling team is equipped to make data-informed decisions. If drilling performance aligns with predicted behavior, operations proceed under the current parameter regime. If deviations are observed—such as unexpectedly low ROP in soft formations or inefficient WOB usage in harder formations—the system prompts a decision-making check.

Three potential scenarios are evaluated:

- **Drilling Parameters are Suboptimal:** If drilling in high UCS zone, formation resistance is suspected to be the primary factor impairing performance, adjustments to WOB, RPM, or flow rate are considered to preserve bit health.
- **Bit May Be Worn Out:** If increased WOB fails to improve ROP in low UCS zones, it suggests diminished bit efficiency, prompting a planned bit trip.
- **Geosteering or Inclination Correction Needed:** If UCS variations suggest an unexpected lithological boundary, the decision tree includes consulting the geosteering team to reassess trajectory control.

For example, in high UCS zones where continued high WOB yields little or no ROP gain, engineers reduce WOB to preserve bit integrity, accepting a slower rate of penetration in exchange for longer bit life and fewer unplanned trips. Conversely, in low UCS zones, failure to improve ROP with additional WOB may signal that the bit is no longer cutting effectively, leading to a decision to pull the bit and replace it.

This closed-loop feedback system ensures that decisions are always grounded in up-to-date geological understanding and equipment response, minimizing non-productive time (NPT), reducing the number of bit runs, and extending bit life while maximizing drilling efficiency.

Field Implementation & Results

The machine learning-powered UCS prediction workflow was piloted in a sandstone-dominated gas field located in the Eastern Province. The region is known for its stratigraphic complexity, interbedded abrasive zones, and varying rock strengths, making it an ideal environment to validate the value of real-time UCS estimation in drilling operations. The objective of this implementation was to test the system's ability to provide continuous UCS feedback, inform drilling decisions, and ultimately improve drilling efficiency and equipment utilization.

Real-Time Integration and Operational Deployment

The UCS workflow was deployed during the drilling of a lateral section. At the core of the system was a Random Forest regression model, trained on historical datasets from several offset wells in the same formation. These wells included direct measurements of compressional slowness (DTC), allowing the model to learn the relationship between standard LWD inputs—gamma ray (GR), bulk density (RHOB), and thermal neutron porosity—and sonic travel time. The model was further validated on wells excluded from the training process, ensuring generalizability to new wellbores.

Once deployed, the model began predicting DTC values in real time as LWD data became available. These predictions were then converted to UCS values using McNally's empirical equation, where DTC is expressed in microseconds per foot and UCS in psi. This equation, while simple, has demonstrated broad applicability in sandstone formations and was chosen for its operational robustness and compatibility with real-time systems.

Lastly, the delivery of these UCS predictions was facilitated through an autonomous petrophysical platform, which interfaced directly with the rig's data acquisition systems. This allowed for seamless visualization and integration of the UCS estimates into the existing real-time monitoring environment used by the drilling team.

Impact on Drilling Decision-Making

The primary aim of delivering UCS data in real time was to improve drilling decisions—specifically, to allow engineers to make timely adjustments to drilling parameters such as Weight on Bit (WOB), RPM, and mud flow rate. The rationale was simple: by having a better understanding of the mechanical strength of the formation being drilled, the team could reduce inefficiencies and preempt bit damage.

For example, in high-UCS zones where the rock is extremely resistant, aggressively applying WOB typically results in rapid bit wear with minimal gain in ROP. In such cases, lowering WOB can help preserve the bit's cutting structure, improving its lifespan without significantly compromising drilling progress. Conversely, in softer, low-UCS formations, if ROP does not increase despite higher WOB, this often indicates that the bit has worn out and needs to be replaced. These types of insights—normally only available in hindsight—were now being inferred on the fly.

The real-time UCS feedback enabled a closed-loop decision process. Drilling parameters were continuously evaluated against predicted UCS and measured ROP. If performance was optimal, the current strategy was maintained. If not, a decision branch was followed—either consulting geosteering teams for possible trajectory changes or adjusting drilling parameters to better match formation characteristics.

This real-time, adaptive loop created a proactive drilling environment where inefficiencies were addressed as they emerged, rather than after they had already impacted performance.

Field Performance: A Comparative Success

The proof-of-concept was executed on a lateral section traditionally viewed as problematic due to its heterogeneity and abrasive intervals. In this first implementation, the section was completed using only three bits, compared to five to seven bits typically required for similar lengths in nearby offset wells. This

represents a 40% improvement in bit utilization, driven not by changes in hardware but by smarter, more responsive drilling behavior enabled by AI-generated UCS data.

In terms of rate of penetration (ROP), the average during the lateral section ranged between 6 to 8 feet per hour, compared to a baseline of approximately 5 feet per hour observed in offset wells. This improvement, though seemingly modest, translates to significant rig time savings over the course of a lateral section, and—more critically—demonstrates that drilling more intelligently does not require sacrificing performance for bit longevity. The workflow allowed for a balanced optimization of both.

Following this initial success, the tool was deployed across seven additional wells in the same field. Consistency in performance was observed across all cases: bit life was extended, ROP was improved or maintained, and the number of unplanned trips was reduced. These deployments reinforced the reliability and adaptability of the machine learning model, as it performed well across wells it had not seen during training, indicating good model generalization across the geological domain.

Operational and Financial Outcomes

Several key operational and economic benefits emerged from the implementation of this AI-powered UCS monitoring system:

- **More Informed Drilling Decisions:** Engineers had a deeper understanding of real-time subsurface behavior, enabling more strategic control of the drilling process.
- **Timely Adjustments:** Instead of relying on delayed post-well analysis, decisions were made proactively, reducing exposure to drilling hazards.
- **Improved ROP:** Higher or sustained rates of penetration were achieved, even in zones where lower ROP is traditionally expected.
- **Extended Bit Life:** Optimized drilling parameters led to fewer bit changes, resulting in reduced NPT and associated rig downtime.
- **Lower Equipment Costs:** With fewer bits used and less unproductive time spent tripping for worn tools, equipment-related expenses were significantly reduced.
- **Rig Time Savings:** Increased drilling efficiency helped shorten the overall well delivery time, freeing up resources for subsequent operations.

These outcomes demonstrate the tangible value of incorporating real-time AI models into drilling workflows. What makes this workflow particularly scalable is its reliance on standard LWD inputs and surface drilling parameters—data already collected in virtually every drilling operation. This eliminates the need for new tools or sensors, allowing for straightforward integration into existing digital infrastructures.

Conclusion

This study presented a practical, real-time workflow for predicting Unconfined Compressive Strength (UCS) from standard Logging-While-Drilling (LWD) data using a machine learning approach. By leveraging a Random Forest model trained on nearby well data, the system enabled continuous prediction of sonic slowness (DTC), which was then converted into UCS estimates through an established empirical relationship. Crucially, the workflow required no additional hardware, instead utilizing routinely collected surface and downhole measurements, making it highly cost-effective and easily deployable across drilling operations.

Field implementation in a sandstone gas reservoir demonstrated the operational value of integrating real-time UCS predictions into drilling decision-making. Engineers were able to dynamically adjust drilling parameters in response to formation strength, optimizing for both rate of penetration (ROP) and bit longevity. The results were tangible: reduced bit usage, improved ROP, fewer trips, and ultimately, better

time and cost efficiency across multiple wells. The system shifted decision-making from reactive to proactive, enabling a higher level of drilling intelligence.

Beyond immediate operational gains, this workflow lays the foundation for more adaptive, data-driven drilling systems. As digital infrastructure across the industry continues to mature, such intelligent workflows can evolve into autonomous decision-making engines that support geosteering, bit wear forecasting, and broader wellbore optimization in real time.

In an industry where marginal gains in performance can lead to substantial reductions in non-productive time and operational risk, the ability to continuously understand and respond to rock strength while drilling represents a critical advancement. The integration of AI in this capacity not only optimizes well delivery but also contributes to a safer, more efficient, and more sustainable future for drilling operations.

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